

Closed-Incision Negative Pressure Wound Therapy Does Not Reduce Surgical Site Infection Rates When Compared to Conventional Wound Dressings in Ankle Fractures Operated via the Lateral or Posterolateral Approach

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Abstract

Aims: To compare the incidence of SSI in ankle fracture patients treated with open reduction and internal fixation, followed by application of either closed-incision negative pressure wound therapy (ciNPWT) or conventional wound dressing (CWD).

Methods: This retrospective cohort study included adults (≥ 18 years) who underwent ankle fracture surgery via the lateral or posterolateral approach between October 2021 and September 2022 (CWD group) and between January 2024 and December 2024 (ciNPWT group). CWD or ciNPWT were applied directly to the skin following wound closure. Data was retrieved through retrospective chart review. The primary outcome was SSI within 6 weeks. SSI risk was calculated with multivariable logistic regression adjusted for body mass index (BMI), American society of anaesthesiologist (ASA) class, current tobacco use, current alcohol consumption and duration of surgery, following confounder evaluation.

Results: A total of 295 patients (median age 58; 65% female) were included, of whom 48% were treated with ciNPWT. SSIs occurred in 15% of patients treated with ciNPWT (12% superficial, 2.8% deep) and in 9.9% treated with CWDs (5.3% superficial, 4.6% deep). Multivariable analysis showed 54% higher odds of SSI with ciNPWT compared to CWD, although not statistically significant (95% CI 0.71–3.38; $p = 0.28$). Odds of deep SSI were 35% lower in the ciNPWT group, again not statistically significant (95% CI 0.13–2.57; $p = 0.55$).

Conclusion: We found no association between the use of ciNPWT after ankle fracture surgery and reduced SSI rates compared with CWDs. Although deep infections were less frequent in the ciNPWT group, the overall infection rate was higher. Prospective, randomized studies are warranted to further clarify the potential benefits of ciNPWT in this patient population.

Introduction

Surgical site infection (SSI) is a common complication following ankle fracture surgery, with a reported incidence of 10-14%¹⁻⁴. Risk factors for SSI are patient-, injury- and surgical incision related⁵. Patient-related risk factors include advanced age, diabetes mellitus, current tobacco use, American Society of Anesthesiologists (ASA) score ≥ 3 , and obesity²⁻⁴. Injury- and surgical incision related risk factors include open fractures, soft-tissue trauma with significant damage to ligaments, tendons, muscles, skin and blood vessels, high-tension incision line, edema, prolonged duration of surgery, and emergency procedures^{5,6}. SSIs are associated with prolonged morbidity, delayed rehabilitation, repeated surgical interventions, and increased healthcare costs^{2,7}. Strategies to reduce SSI rates are therefore clinically meaningful.

To reduce SSI rates, closed-incision negative pressure wound therapy (ciNPWT) has been suggested as an alternative to conventional wound dressings (CWD). Multiple studies have shown that ciNPWT can reduce SSI rates^{5,8,9}. A 2020 systematic review on randomized controlled trials found a 39% reduction across a wide range of surgical procedures, including surgery of lower limb fractures, total hip and knee arthroplasty, caesarean section, laparotomy, thoracotomy, mammoplasty, vascular bypass and revascularization procedures, when ciNPWT was used instead of CWD⁷. The application of continuous negative pressure, typically -125 mmHg, is reported to remove exudate and oedema, reducing bacterial contamination, and enhancing perfusion of blood flow and cell proliferation, creating an environment conducive to woundhealing^{5,10-13}. Although several studies suggest that ciNPWT may reduce SSI risk in various surgical indications, evidence is inconclusive^{7,14-17}.

Because ciNPWT is relatively costly, its use is typically limited to high-risk procedures or patients at high risk of developing SSI^{5,16}. There is moderate-certainty evidence that ciNPWT in surgery for lower limb fracture is not cost effective¹⁰. To date, no studies have investigated ciNPWT in ankle fracture surgery - a procedure associated with a relatively high incidence of SSI⁵.

Since October 2023, ciNPWT (Avance® Solo, Mölnlycke, Sweden) has been routinely used on lateral and posterolateral incisions in ankle fracture surgery at our university level II trauma centre in Copenhagen, Denmark. Here, we compared the occurrence of SSI in patients treated with ciNPWT vs. CWD. We hypothesized that ciNPWT decreases the risk of SSI following ankle fracture surgery via the lateral or posterolateral approach.

Methods

Study Design and Population

This retrospective cohort study included adults (≥ 18) who underwent ankle fracture surgery via the lateral or posterolateral approach. Two consecutive time periods were examined: October 2021 to September 2022 (CWD group) and January to December 2024 (ciNPWT group). Institutional postoperative wound management protocols were adapted to the ciNPWT protocol from October 2023 onwards to potentially reduce SSIs. Patients were identified through the electronic health record system *Sundhedsplatformen* (Epic Systems Inc.). Data were obtained through retrospective chart review¹⁸. Patients were excluded if they (i) did not receive standard prophylactic perioperative antibiotics (2 g cloxacillin or 1.5 g cefuroxime), (ii) had missing documentation of wound dressing or incision site, (iii) were lost to follow-up or (iv) died within six weeks following surgery.

Baseline demographics included sex, age, body mass index (BMI), current tobacco use, alcohol consumption (> 10 units a week), comorbidities, and American society of anaesthesiologist (ASA) class. Fracture characteristics included high or low energy trauma, Gustilo Anderson (GA) classification for open fractures and Arbeitsgemeinschaft für Osteosynthesefragen (AO) classification of malleolar fractures. Surgery characteristics included duration of surgery and surgeons' level of experience, temporary fixation with external fixator, and type of surgery (open reduction and internal fixation (ORIF) with plate osteosynthesis or fibula nail osteosynthesis), and postoperative bandage and antibiotics.

Postoperative wound characteristics included wound dehiscence, redness, serous or purulent drainage, maceration, bullae, visible metal or bone of the medial, lateral and posterolateral incisions, traumatic wounds and dorsal pressure ulcers on the ankle. Wound assessment was supported by chart review, wound photographs, antibiotic prescriptions and microbiology samples, and recorded in the outpatient clinic.

Endpoints

The primary outcome was SSI within 6 weeks post operative, defined by a semiquantitative grading system for wound infections after osteosynthesis, based on The Wound after Osteosynthesis Score (WOK)¹⁹:

0: No complication.

1: Local redness and minimal wound dehiscence or wound drainage that can be treated non-operative without the need for antibiotics.

2: Superficial SSI with grade 1 problems requiring antibiotics.

3: Deep SSI without visible metal or bone requiring antibiotics and surgical debridement.

4: Deep SSI with visible metal or bone requiring implant removal, debridement, and antibiotics.

SSI was defined as grade 2–4, with superficial SSI corresponding to grade 2 and deep SSI to grades 3–4.

Secondary outcomes comprised local debridement and revision surgery including hardware removal, arthrodesis, new osteosynthesis or transfemoral amputation due to SSI of the lateral or posterolateral incision within 1 year.

Adverse events were recorded in the outpatient clinic and included pain, nerve injury, local debridement, revision surgery, and mechanical failure of osteosynthesis within 1 year, due to other complications than SSI of the lateral or posterolateral incision.

Exploratory hypotheses

Two exploratory hypotheses were assessed: (i) We hypothesized a lower overall SSI rate at 6-week follow-up, and (ii) deep SSI in the ciNPWT group vs. CWD within 6 weeks post-surgery.

Postoperative Treatment Protocols

CWD and ciNPWT were applied directly to the skin following wound closure under sterile conditions.

CWD group:

- Conventional dressing (Mepilex® Border Post-Op, Mölnlycke, Sweden), and short leg cast, changed to circular short leg cast within 1 week.
- Cast changed to Walker boot (PROCARE® Nextep™ Contour 2) and class 1 compression stockings at 3 weeks. Removed after 6 weeks.
- Dressing and suture removal at week 3
- Outpatient follow-up at week 3 and 6
- Full weightbearing post-surgery

ciNPWT Group:

- ciNPWT (Avance® Solo, Mölnlycke, Sweden), Coban™ compression bandage and walker boot (PROCARE® Nextep™ Contour 2) for 6 weeks
- Change of ciNPWT plaster after 1 week and ciNPWT removal after 2 weeks
- Dressing and suture removal at week 2
- Outpatient follow-up at week 1, 2 and 6
- Full weightbearing post-surgery

The dressing was typically changed only once, and never more than twice.

Statistics

Wound grades were summarized descriptively. Categorical variables were presented as counts and percentages, and continuous variables as mean (SD) or median (IQR).

Logistic regression was performed with SSI (grade 2–4) within 6 weeks as the dependent variable and treatment group (ciNPWT vs. CWD) as the main exposure. All variables from **Table I** were evaluated individually as potential confounders by adding them to the multivariable model. The final multivariable logistic regression model for the primary outcome was adjusted for BMI, ASA class, current tobacco use, current alcohol consumption (≥ 10 units/week), and duration of surgery.

In addition to the primary analysis, two multivariable logistic regression analyses were performed to evaluate the exploratory hypotheses: (i) a lower overall SSI rate at 6-week follow-

up, and (ii) deep SSI in the ciNPWT group vs. CWD within 6 weeks post-surgery. For these analyses, all potential confounders were evaluated individually, as in the primary analysis, and included as follows: (1) BMI, current tobacco use, current alcohol consumption, postoperative antibiotics, and duration of surgery; and (2) BMI, ASA class, postoperative antibiotics, and type of osteosynthesis. All models used the same primary exposure variable.

Because of a relevant difference in the incidence of pressure ulcers dorsally on the ankle and medial wound complications between groups (more common in the ciNPWT group), these variables were considered potential mediators. To ensure that not all the SSIs were due to the mediators, a multivariable analysis of the primary outcome including the potential mediators was conducted. We also performed a logistic regression excluding the patients with dorsal pressure ulcers and medial wound complications.

Odds ratios (ORs) with 95% confidence intervals (CIs), and p-values were calculated for the logistic- and multivariable logistic regressions. Significance was defined as $p < 0.05$. Model fit was assessed using the Hosmer–Lemeshow goodness of fit test.

Statistical analyses were performed using R (version 4.4.2; R Foundation for Statistical Computing, Vienna, Austria) in RStudio (version 2025.05.0+496).

Power Analysis

A power analysis was performed based on a study of lower limb-fracture surgery by Stannard et al., which reported SSIs in 9.7% of patients treated with ciNPWT and 19% of patients in the control group¹⁴. Assuming a power of 80% and a two-sided significance level of 0.05, 222 patients would be required in each group. Because the sample size was determined by the number of available patients, only 165 patients had been treated with ciNPWT at the time of data collection, the study had approximately 67.5% power to detect a statistically significant difference.

Ethics

The project was approved by the Danish Data Protection Agency (journal no. R-23042748). The study adhered to principles described in the Declaration of Helsinki. Data were stored on a secure server accessible only to the research team.

Results

Study population

A total of 330 patients underwent ankle fracture surgery via the lateral or posterolateral approach during the study period, of whom 295 met the inclusion criteria (**Figure 1**). In the CWD group, 162 patients were identified and 152 were included after exclusions for loss to follow-up (n=8) or death (n=2). In the ciNPWT group, 168 patients were identified and 143 were included, with exclusions due to lack of ciNPWT application (n=18), loss to follow-up (n=5), or death (n=2). All the patients received standard prophylactic perioperative antibiotics, and no patients had missing documentation of wound dressing or incision site.

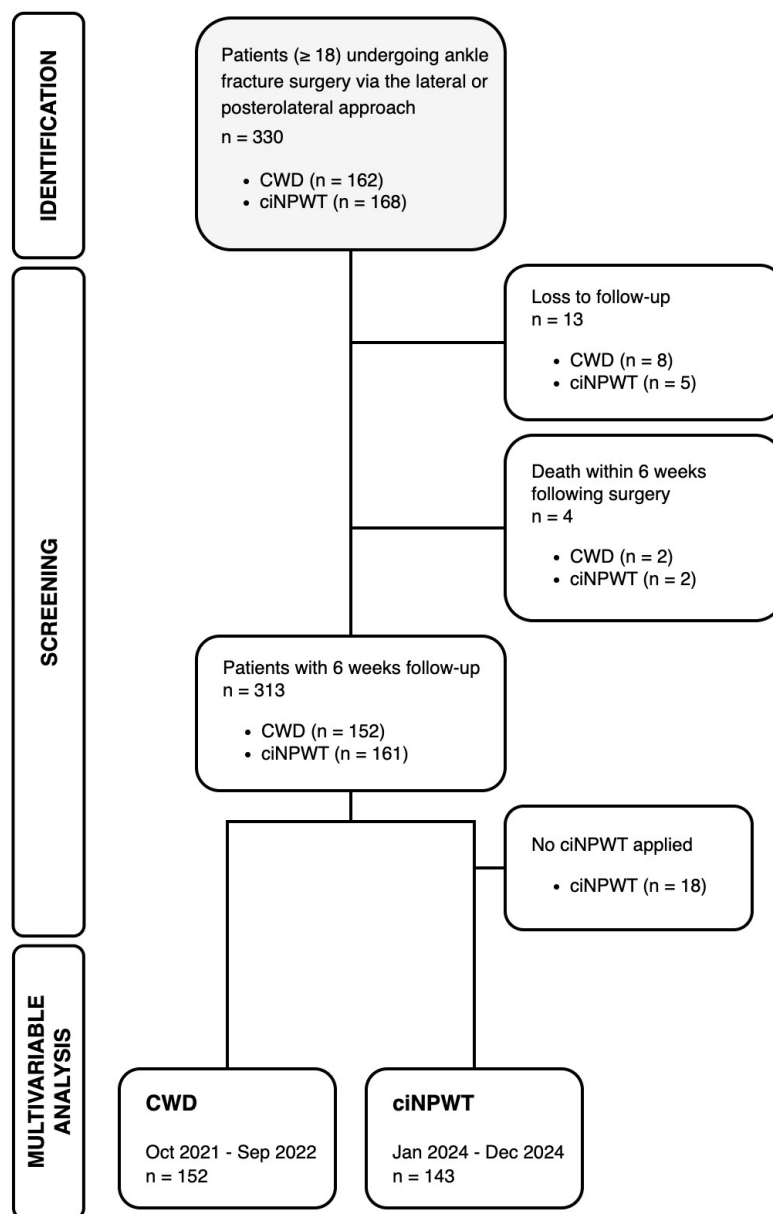


Figure 1. Study design and screening process. Retrospective cohort study with multivariable analysis.

Characteristics were generally comparable between groups, although the CWD group had a slightly higher proportion of patients with diabetes, ASA ≥ 3 , high energy trauma, fibula nail osteosynthesis, postoperative antibiotic use, and shorter duration of surgery (**Table I**). Open fractures occurred in 7% of patients in both groups. The median duration of ciNPWT was 14 days (range 2-19).

Table I. Characteristics. Patient demographic data, comorbidities, injury and surgery characteristics.

Characteristics	Treatment		
	Overall N = 295	ciNPWT N = 143	CWD N = 152
Age (y)¹	58 (44, 72)	57 (38, 71)	61 (45, 73)
Gender²			
Female	191 (65%)	93 (65%)	98 (64%)
Male	104 (35%)	50 (35%)	54 (36%)
BMI (kg/m³)^{3*}	26 (6)	27 (6)	28 (5)
ASA^{2*}			
1	70 (24%)	40 (28%)	30 (20%)
2	161 (55%)	80 (56%)	81 (53%)
3	64 (22%)	23 (16%)	41 (27%)
Current tobacco use^{2*}	75 (26%)	38 (28%)	37 (25%)
Current alcohol consumption (≥ 10 units per week)^{2*}	54 (19%)	27 (20%)	27 (18%)
Diabetes mellitus²	28 (9.5%)	11 (7.7%)	17 (11%)
High energy trauma²	36 (12%)	15 (10%)	21 (14%)
Gustilo-Anderson classification (open fracture)²			
0	275 (93%)	133 (93%)	142 (93%)
1	3 (1.0%)	2 (1.4%)	1 (0.7%)
> 1	17 (5.8%)	8 (5.6%)	9 (5.9%)
AO classification²			
A	2 (0.7%)	1 (0.7%)	1 (0.7%)
B	235 (80%)	117 (82%)	118 (78%)
C	57 (19%)	25 (17%)	32 (21%)
Temporary fixation²	66 (22%)	33 (23%)	33 (22%)
Type of surgery²			
ORIF	191 (65%)	99 (69%)	92 (61%)
Fibula nail	104 (35%)	44 (31%)	60 (39%)
Days to final surgery¹	6 (3, 9)	5 (2, 8)	6 (3, 9)
Duration of surgery (min)^{1*}	117 (92, 114)	121 (95, 152)	115 (87, 138)
Preoperative antibiotics²	36 (12%)	17 (12%)	19 (13%)
Postoperative antibiotics²	65 (22%)	23 (16%)	42 (28%)

¹ Median (Q1, Q3); ² n (%); ³ Mean (SD); *Adjusted for in the multivariable regression for primary outcome

SSIs

Across all included patients, 12.2% developed an SSI within 6 weeks after surgery (superficial 8.5%, deep 3.7%). In the CWD group, 15 patients (9.9%) developed an SSI, of whom 8 (5.3%) had a superficial and 7 (4.6%) a deep SSI. In the ciNPWT group, 21 patients (14.7%) developed an SSI, including 17 (11.9%) superficial and 4 (2.8%) deep infections. Overall, 63 patients (21%) experienced wound grade 1 complications: 21 (14%) in the CWD group and 42 (29%) in the ciNPWT group.

Table II. Incidence of SSI according to wound grades.

Infection type	Wound grade	Treatment		
		Overall N = 295	ciNPWT N = 143	CWD N = 152
No Infection	0-1	259 (87.8)	122 (85.3%)	137 (90.1%)
	0	196 (66%)	80 (56%)	116 (76%)
	1	63 (21%)	42 (29%)	21 (14%)
Infection	2-4	36 (12.2%)	21 (14.7%)	15 (9.9%)
Superficial	2	25 (8.5%)	17 (11.9%)	8 (5.3%)
Deep	3-4	11 (3.7%)	4 (2.8%)	7 (4.6%)
	3	4 (1.4%)	3 (2.1%)	1 (0.7%)
	4	7 (2.4%)	1 (0.7%)	6 (3.9%)

n (%)

Unadjusted logistic regression showed a non-significant increase in the odds of SSI in the ciNPWT group compared with the CWD group (OR 1.57, 95% CI 0.78–3.24; $p = 0.21$). After adjustment for relevant confounders, the association remained non-significant (adjusted OR 1.54, 95% CI 0.71–3.38; $p = 0.28$).

The exploratory hypotheses showed: (i) no significant difference of overall SSI rate at 6-week follow-up (OR 1.11, 95% CI 0.37–3.29; $p = 0.85$), and (ii) approximately one-third lower odds ratio for deep infection in the ciNPWT group vs. CWD within 6 weeks post-surgery, although not significant (OR 0.65, 95% CI 0.13–2.57; $p = 0.55$).

Secondary outcomes and adverse events

Secondary outcomes are listed in **Table III**. Local debridement due to SSI of the lateral or posterolateral incision was performed in 7.2% of patients treated with CWD and 9.8% treated with ciNPWT. Surgical debridement, including hardware removal, was performed in 3.6% and 1.4%, and respectively. One patient in each group underwent reoperation with arthrodesis, and 1 patient in the CWD group underwent a transfemoral amputation due to SSI.

Among adverse events (**Table IV**), dorsal pressure ulcers at the ankle and medial wound complications were substantially more frequent in the ciNPWT group (29% vs 1.3% and 24% vs 11%, respectively). Many of the adverse events were due to infections following dorsal pressure ulcers and medial wounds, but also due to hardware prominence and hardware failure. Five patients underwent surgical debridement in the CWD group and 3 in the ciNPWT group. Two in each group were reoperated with arthrodesis.

Table III. Secondary outcomes, surgery due to SSI of the lateral or posterolateral incision within 1 year.

Secondary outcomes	Treatment		
	Overall	ciNPWT	CWD
	N = 295	N = 143	N = 152
Local debridement (not surgery)	25 (8.5%)	14 (9.8%)	11 (7.2%)
Surgical debridement (including hardware removal)	6 (2.0%)	2 (1.4%)	4 (3.6%)
Arthrodesis	2 (0.7%)	1 (0.7%)	1 (0.7%)
Transfemoral amputation	1 (0.3%)	0 (0%)	1 (0.7%)
n (%)			

Table IV. Adverse events (not due to SSIs of the lateral or posterolateral incision) within 1 year.

Treatment

Adverse events	Overall	ciNPWT	CWD
	N = 295	N = 143	N = 152
Pain	57 (19%)	18 (13%)	39 (26%)
Nerve injury	39 (13%)	16 (11%)	23 (15%)
Local debridement (no open surgery)	19 (6.4%)	11 (7.7%)	8 (5.3%)
Surgical debridement (including hardware removal)	8 (2.7%)	3 (2.1%)	5 (3.3%)
Mechanical failure	15 (5.3%)	9 (6.5%)	6 (4.1%)
Arthrodesis	4 (1.3%)	2 (1.4%)	2 (1.3%)
Dorsal pressure ulcer	43 (15%)	41 (29%)	2 (1.3%)
Medial wound complication	50 (17%)	34 (24%)	16 (11%)
Transfemoral amputation due to infected pressure ulcer	1 (0.3%)	1 (0.7%)	0 (0%)
n (%)			

Mediators

Patients who developed dorsal pressure ulcers had a twofold higher incidence of SSI of the lateral or posterolateral incision (21% vs. 11%) than those without. Those with medial wound complications had a threefold higher incidence of SSI of the lateral or posterolateral incision (28% vs. 9%) than those without, suggesting that these complications may have mediated part of the observed effect ciNPWT had on SSI.

Inclusion of the proposed mediators in the multivariable model attenuated the association between ciNPWT and SSI, with the odds ratio decreasing from 1.54 to 1.09 (95% CI 0.44 to 2.62; $p = 0.85$). This suggests that part of the observed association may be mediated through these complications, although the effect was not fully accounted for.

In a secondary analysis, cases with dorsal pressure ulcers and/or medial wound complications were excluded, resulting in a further reduction of the odds ratio to 1.02, but not significant (95% CI 0.30 to 3.13; $p = 0.97$). However, this approach alters the composition of the study population and may affect comparability between the ciNPWT and CWD groups, as these complications may themselves be influenced by the intervention. Consequently, this analysis no longer reflects the original comparison of the full cohort and is susceptible to selection bias.

These findings should therefore be interpreted with caution. While they are consistent with the hypothesis that these complications contribute to the observed association between ciNPWT and SSI, they do not establish mediation, and the primary analysis of the full cohort remains the most appropriate estimate of the overall effect.

Characteristic	Overall N = 295 [†]	Treatment Received	
		ciNPWT N = 143 [†]	CWD N = 152 [†]
Pain	57 (19%)	18 (13%)	39 (26%)
Nerve damage	39 (13%)	16 (11%)	23 (15%)
Debridement (not surgery)	44 (15%)	25 (17%)	19 (13%)
Revision surgery	8 (2.7%)	2 (1.4%)	6 (3.9%)
Mechanical failure	15 (5.3%)	9 (6.5%)	6 (4.1%)
Reoperation	28 (9.5%)	9 (6.3%)	19 (13%)
[†] n (%)			

Discussion

In this retrospective cohort study, we found no significant reduction in SSI among ankle fracture patients treated with ciNPWT following ankle fracture surgery when compared with CWD. Although the ciNPWT group demonstrated fewer deep infections, the overall SSI rate was higher, and differences were not statistically significant.

Our results are in line with previous findings of ciNPWT in orthopaedic trauma surgery. A randomized controlled study (RCT) by Christ et al. found no significant advantage of ciNPWT over standard dressings in patients with acetabular fractures and even observed higher SSI rates in the ciNPWT group (15.2% vs. 6.1%)¹⁵. Similarly, an RCT by Costa et al., including 1,548 patients, found no significant difference in deep SSI (5.8% vs. 6.7%) when using ciNPWT (PICO®, Smith & Nephew, US) following orthopaedic trauma surgery of the acetabulum, hip, femur, patella, tibia, fibula, and foot. Further, ciNPWT did not reduce treatment costs and was associated with a low probability of cost-effectiveness¹⁶.

Several randomized trials and systematic reviews have however reported reductions in SSI following ciNPWT across diverse surgeries, including total hip arthroplasty (THA), total knee arthroplasty (TKA), tibia plateau, pilon, and calcaneus fracture surgery. Yet, study population, fracture type, mechanism of injury, ciNPWT systems used, pressure settings, and duration of therapy differ from our study^{5,7,9,14,17}. Stannard et al. reported a reduction in infection rates from 18.9% to 9.9% in tibial plateau, pilon, and calcaneal fractures—typically high-energy injuries—making comparisons to ankle fractures difficult¹⁴. A prospective cohort study by Dingemans et al. evaluated a portable ciNPWT system (PICO®, Smith & Nephew, US) used prophylactically after major foot and ankle surgery. Although the results were not statistically significant, the study reported a decrease in SSI from 14.9% to 4.3% with ciNPWT¹⁷. In contrast to our study, this cohort excluded open fractures, used a substantially lower negative

pressure setting (80 mmHg vs. 125 mmHg with Avance® Solo), and applied ciNPWT for a much shorter duration (mean 2.5 days vs. median of 14 days in our study). The cohort included a wide range of lower-extremity fractures with varied incision types, but only ten cases involved ankle fractures treated via a lateral approach.

Finally, a meta-analysis by Shiroky et al. demonstrated a significant overall reduction in SSIs with ciNPWT, covering multiple specialties and ciNPWT systems. Their estimated number needed to treat (NNT) of approximately 100 patients with clean wounds to prevent one SSI highlights the modest absolute benefit⁷. In contrast to our findings, both Stannard et al. (16,5 to 8,6%) and Shirokoy et al. (RR 0.76) demonstrated a decrease in wound dehiscence in patients treated with ciNPWT^{7,14}.

A notable finding in our study was the substantially higher incidence of dorsal pressure ulcers (29% vs. 1%) and medial wound complications (24% vs. 10%) in the ciNPWT compared to the CWD group. These complications were likely not caused by the ciNPWT itself—which was applied only to the lateral incision—but potentially by the use of elastic compression bandaging and controlled ankle motion (CAM) boot instead of a cast in the ciNPWT group. Although dorsal pressure ulcers and medial wound complications had no direct effect on the lateral incision, they may have contributed to increased swelling, discomfort, or local inflammation, potentially affecting wound healing. The additional analysis investigating their potential mediating effect may have introduced overadjustment or selection bias, and these findings should be interpreted with caution. The effect of ciNPWT on SSI appears to extend beyond these complications. For future studies, postoperative interventions should avoid simultaneous modifications of multiple protocol elements, including changes to compression bandaging, as these factors may have contributed to the increased incidence of pressure ulcers and an associated risk of infection.

Because the sample size was determined by the number of available patients, this study was underpowered, with an estimated 67.5% probability of detecting a clinically meaningful effect. Consequently, the risk of a type II error was high, meaning that a true beneficial effect of ciNPWT could have remained undetected²⁰. In addition, the exploratory analyses—including SSI at 6 weeks and deep SSI within 6 weeks—further increase the risk of type I error due to multiple testing, although none of these analyses reached statistical significance²¹.

Despite these limitations, the study's strengths include a relatively large sample size for this surgical population and the use of 2 cohorts treated at the same institution, reducing variability

in surgical protocols, documentation, and postoperative pathways. Compared with previous studies, the use of higher negative pressure and longer ciNPWT application duration in this study would be expected to increase the likelihood of detecting a beneficial effect. Despite a greater burden of comorbidities in the CWD cohort, this group demonstrated a lower incidence of SSI, however not statistically significant. Overall, our findings suggest that ciNPWT does not confer a clear advantage in ankle fracture patients operated via a lateral or posterolateral incision.

Conclusion

In this retrospective cohort study, we did not find that ciNPWT reduced SSI rates after ankle fracture surgery via a lateral or posterolateral incision compared with conventional dressings. Although deep infections were less frequent in the ciNPWT group, the overall SSI rate was higher. Prospective, randomized studies are warranted to determine the true effect of ciNPWT in ankle fracture surgery.

Conflicts of interest

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References

1. **SooHoo NF, Krenek L, Eagan MJ, Gurbani B, Ko CY, Zingmond DS.** Complication Rates Following Open Reduction and Internal Fixation of Ankle Fractures: *J Bone Jt Surg-Am Vol* 2009;91(5):1042–1049.
2. **Bergström J, Möller Rydberg E, Wennergren D, Svensson Malchau K.** Incidence and Risk Factors for Surgical Site Infection in Ankle Fractures: An Observational Study of 480 Patients in Sweden. *J Clin Med* 2023;12(20):6464.
3. **Olsen LL, Møller AM, Brorson S, Hasselager RB, Sort R.** The impact of lifestyle risk factors on the rate of infection after surgery for a fracture of the ankle. *Bone Jt J* 2017;99-B(2):225–230.
4. **Ovaska MT, Mäkinen TJ, Madanat R, Vahlberg T, Hirvensalo E, Lindahl J.** Predictors of poor outcomes following deep infection after internal fixation of ankle fractures. *Injury* 2013;44(7):1002–1006.
5. **Willy C, Agarwal A, Andersen CA, Santis GD, Gabriel A, Grauhan O, et al.** Closed incision negative pressure therapy: international multidisciplinary consensus recommendations. *Int Wound J* 2016;14(2):385–398.
6. **Bazarov I, Kim J, Richey JM, Dickinson JD, Hamilton GA.** Minimally Invasive Plate Osteosynthesis for Treatment of Ankle Fractures in High-Risk Patients. *J Foot Ankle Surg* 2018;57(3):494–500.
7. **Shiroky J, Lillie E, Muaddi H, Seigny M, Choi WJ, Karanicolas PJ.** The impact of negative pressure wound therapy for closed surgical incisions on surgical site infection: A systematic review and meta-analysis. *Surgery* 2020;167(6):1001–1009.
8. **Dyck BA, Bailey CS, Steyn C, Petrakis J, Urquhart JC, Raj R, et al.** Use of incisional vacuum-assisted closure in the prevention of postoperative infection in high-risk patients who underwent spine surgery: a proof-of-concept study. *J Neurosurg Spine* 2019;31(3):430–439.
9. **Kim J-H, Kim H-J, Lee D-H.** Comparison of the Efficacy Between Closed Incisional Negative-Pressure Wound Therapy and Conventional Wound Management After Total

Hip and Knee Arthroplasties: A Systematic Review and Meta-Analysis. *J Arthroplasty* 2019;34(11):2804–2814.

10. **Norman G, Shi C, Goh EL, Murphy EM, Reid A, Chiverton L, et al.** Negative pressure wound therapy for surgical wounds healing by primary closure. Cochrane Wounds Group, ed. *Cochrane Database Syst Rev* [Internet] 2022 [cited 22 Dec 2025];2022(4).
11. **Normandin S, Safran T, Winocour S, Chu CK, Vorstenbosch J, Murphy AM, et al.** Negative Pressure Wound Therapy: Mechanism of Action and Clinical Applications. *Semin Plast Surg* 2021;35(3):164–170.
12. **Timmers MS, Le Cessie S, Banwell P, Jukema GN.** The Effects of Varying Degrees of Pressure Delivered by Negative-Pressure Wound Therapy on Skin Perfusion: *Ann Plast Surg* 2005;55(6):665–671.
13. **Mendonca DA, Papini R, Price PE.** Negative-pressure wound therapy: a snapshot of the evidence. *Int Wound J* 2006;3(4):261–271.
14. **Stannard JP, Volgas DA, McGwin G, Stewart RL, Obremskey W, Moore T, et al.** Incisional Negative Pressure Wound Therapy After High-Risk Lower Extremity Fractures. *J Orthop Trauma* 2012;26(1):37–42.
15. **Crist BD, Oladeji LO, Khazzam M, Della Rocca GJ, Murtha YM, Stannard JP.** Role of acute negative pressure wound therapy over primarily closed surgical incisions in acetabular fracture ORIF: A prospective randomized trial. *Injury* 2017;48(7):1518–1521.
16. **Costa ML, Achten J, Knight R, Png ME, Bruce J, Dutton S, et al.** Negative-pressure wound therapy compared with standard dressings following surgical treatment of major trauma to the lower limb: the WHiST RCT. *Health Technol Assess* 2020;24(38):1–86.
17. **Dingemans SA, Birnie MFN, Backes M, De Jong VM, Luitse JS, Carel Goslings J, et al.** Prophylactic negative pressure wound therapy after lower extremity fracture surgery: a pilot study. *Int Orthop* 2018;42(4):747–753.
18. **Vassar M, Matthew H.** The retrospective chart review: important methodological considerations. *J Educ Eval Health Prof* 2013;10:12.

19. **Jensen NM, Hansen AØ, Thonsgaard S, Simony A, Cavallius C, Østergaard RS, et al.** Assessment of ankle fracture surgical wounds: the development and testing of the Wound after Osteosynthesis Kolding (WOK) score. *Injury* 2025;56(6):112345.
20. **Banerjee A, Chitnis U, Jadhav S, Bhawalkar J, Chaudhury S.** Hypothesis testing, type I and type II errors. *Ind Psychiatry J* 2009;18(2):127.
21. **Gerke O, Høilund-Carlsen PF, Vach W.** Forskellig betydning af p-værdien i eksplorerende og konfirmerende hypotesetestning.